

VALIDATION STUDY: TURBULENT CHANNEL FLOW FOR FRICTION REYNOLDS NUMBER OF 180

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ABSTRACT

A validation study of selected turbulence statistics from a DNS by Moser et al., (1999) is performed. Turbulent channel flow at a friction Reynolds number of 180 is simulated through OpenFOAM software. An elliptic-blending Reynolds-stress model for turbulent flow is utilized with the simpleFoam solver. Uncertainty is quantified and OpenFOAM simulation results are compared to the DNS results at the corresponding Reynolds number. Agreements and discrepancies between the results are discussed to ultimately make a statement on the model error.

Keywords: OpenFOAM, EBRSM, Channel flow

NOMENCLATURE

Re_τ	friction Reynolds number (-)
u_τ	friction velocity $\left(\frac{m}{s}\right)$
ε	dissipation $\left(\frac{m^2}{s^3}\right)$
R_{ii}	normal stress components $\left(\frac{m^2}{s^2}\right)$
R_{ij}	shear stress components $\left(\frac{m^2}{s^2}\right)$
U^+	streamwise dimensionless velocity (-)
y^+	Wall unit (-)

1. INTRODUCTION

Fully developed turbulent channel flow has been thoroughly researched through direct numerical simulations (DNS), Reynolds averaged numerical simulations (RANS), and experimental setups for insight into wall-bounded turbulence. This type of flow has simple geometry with intuitive meshing algorithms, which makes it ideal for simulating wall-bounded turbulence through computational modelling. The main difference between the various studies that have been completed over the years are their simulation type and Reynolds number, both of which are capable of significantly influencing the results of computed turbulence statistics.

The purpose of this study is to dissect one of these computational models, a DNS by Moser et al., (1999), to validate a separate computational model, RANS modelling via Open Field Operation And Manipulation (OpenFOAM), at the same friction Reynolds number, $Re_\tau = 180$. Comparing at the same friction Reynolds number will ultimately provide an idea of

model error between the two simulation types. For the purposes of quantifying uncertainty and model error for the OpenFOAM RANS model, the DNS will be treated as “truth”, as no uncertainties in the referenced paper are provided.

2. MATERIALS AND METHODS

Computational materials were used to perform this study. These include a laptop with 24 cores, a high power cluster (provided by Texas A&M), the AGARD database for DNS results, OpenFOAM version 2412 for modelling, and python for post processing. Further information for the setup of both the DNS (for comparison purposes) and the RANS model is provided in sections 2.1 and 2.2, respectively.

2.1 Direct Numerical Simulation Model

The work by Moser et al., (1999) was a direct numerical simulation of turbulent channel flow for various friction Reynolds numbers, $Re_\tau = 180, 395, \text{ and } 590$. A Chebychev-tau formulation was implemented in the y direction (normal to the wall) and a Fourier representation in the x and z directions (streamwise and spanwise, respectively). Periodic boundary conditions are held in the x and z directions. Though the mesh structure is not specified, it is stated that the cell domain has 128 cells in the x and z directions, and 129 cells in the y direction. The physical geometry of the domain is $4\pi\delta \times \frac{4}{3}\pi\delta \times 2\delta$ in the x, y, z directions, respectively, where δ is the channel half-width. For $Re_\tau = 180$, initialization data was taken from Moser et al., (1987).

2.2 OpenFOAM Model

The OpenFOAM model follows the “planeChannel” tutorial case from the OpenFOAM repository, which served as the foundation for the configuration used in this study. All the initial conditions were corrected to the appropriate friction Reynolds number, and various systems and controls were adjusted to avoid divergence. This case utilizes the simpleFoam solver with an Elliptic Blending Reynolds Stress Model (EBRSM) for turbulence simulation. This turbulence model was selected due to its high accuracy at low friction Reynolds numbers (far surpassing that of kOmegaSST or kEpsilon) and its excellent near-wall treatment, though at an increased computational cost. The computational domain was constructed using blockMesh

with a nominal $10 \times 80 \times 1$ cells in the streamwise (x), wall-normal (y), and spanwise (z) directions, respectively, and physical dimensions of $1 \times 2 \times 1$ in x, y, z. This geometry is shown in Figure 1. Geometric stretching was applied in the wall-normal direction to adequately resolve the viscous sublayer. Although many (over 10) were tested, three mesh resolutions were used ($N = 20, 40, 80$ cells per half-channel block) in the y direction to assess grid convergence, and the effective cell height, h_{eff} , was calculated for each grid to support a Grid Convergence Index (GCI) analysis.

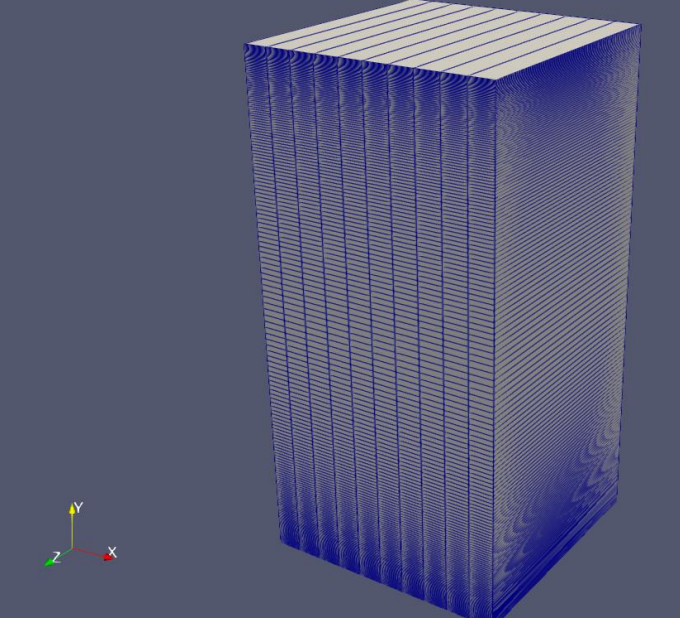


FIGURE 1: Geometry used to simulate channel flow in OpenFOAM

Boundary conditions replicated the DNS setup as closely as possible, having periodicity (cyclic) in the x and z directions and no-slip walls at the top and bottom of the channel. The solver ran until a fully developed flow was achieved, monitoring wall shear stress and mean velocity profiles for convergence. A python postprocessing script was developed to speed up data extraction and automate plotting. Time-averaged statistics were extracted for the mean velocity, turbulent kinetic energy, dissipation rate, production rate, and Reynolds stress components at each grid resolution to be compared to DNS data. For the purposes of finding the observed order of accuracy, only the selected Reynolds stress components (R_{uu} , R_{uv} , R_{vv} , and R_{ww}) and u^+ are reported for quantifying discretization uncertainty, though all simulation variables go through the calculations.

2.3 Model Discrepancies

The physical domain and cell structures are notably different between the two models in their setup. These dimensions differ from the DNS due to time constraints, as simulating 2 million cells took an extreme amount of computational time and power, even with optimized features such as mesh renumbering and parallel simulations. Additionally, the planeChannel case from the OpenFOAM repository was used for a channel flow

validation study for a higher $Re_\tau = 395$. The same geometry and boundary treatment was adopted to reproduce the validation approach and to limit computational expense, as the results from the $Re_\tau = 395$ case were very promising. This configuration produces mean profiles that are predominantly functions of y^+ and thus enables efficient assessment of near-wall statistics. It is noted that suppressing the spanwise direction reduces the ability to generate full three-dimensional turbulent structures; consequently, Reynolds stress components that rely on spanwise fluctuations are expected to be affected. Therefore, comparisons of higher-order stress components (R_{uu} , R_{uv} , R_{vv} , R_{ww}) are made with care, and discretization uncertainty and non-monotonic convergence are quantified where appropriate.

Many computational challenges were encountered when running the simulations. The most common of which was dissipation diverging. It should be noted that EBRSM models are extremely sensitive to initial conditions, so much so that OpenFOAM implemented a script to automatically fill in these fields to avoid such disasters (setTurbulenceFields). Many combinations of boundary conditions, solvers, and tools were tested until a very sensitive but finally stable case was able to be constructed for such a low Re . For the interested reader who wishes to run the models detailed in this paper, the directory “proj” contains the main running case, “case”. Inside this directory is a bashscript “run” which will automate the running of a case for $N = 80$ cells and automatically post process. Another directory, “data” contains the python script and data extracted from such runs to perform analysis.

3. RESULTS AND DISCUSSION

3.1 Simulation Inputs

The basis for all initial conditions and model parameters ultimately stem from the friction Reynolds number. From Moser et al. (1987), the relationship between the mean bulk velocity, U_b and the friction velocity are given by Equation 1.

$$Re_\tau = \frac{U_b}{u_\tau} = 15.63 \quad (1)$$

By keeping the friction velocity as 1 m/s, the mean velocity can simply be put as 15.63 m/s. A channel half width of 1 m was chosen for simplicity, allowing the kinematic viscosity, ν , to simply be $1/Re_\tau$. Dissipation and turbulent kinetic energy were acquired from the DNS results, with the wall boundary values corresponding to $y = 0$ m and the internal field initial guess at the centerline. Turbulent viscosity and dissipation were also given a wall function with a low- Re corrector, to aid in the EBRSM wall-treatment. Zero-gradient boundary conditions were applied at the walls for pressure.

3.1 Simulation Results

The model ran for 100,000 seconds or until a convergence of $1.0\text{e-}11$ was reached by pressure, velocity, and Reynolds stress terms. Results of the model simulations are shown in Figures 2 through 6. The comparison error from results and the DNS benchmark data are highlighted in Figure 7 and Tables 1 and 2, showing the (absolute) pointwise, L_2 , and L_∞ norm of the error. Overall, there is good agreement between the DNS and the OpenFOAM simulation. All turbulence statistics follow the expected profiles, though sometimes at different magnitudes than the DNS. Data tends to match closer at the wall than at the peak, which is expected as the EBRSM excels in near-wall simulations. At a glance, R_{ww} plot has the most glaring relative difference between the benchmark data. The reason for this is suggested to be due to two-dimensional nature of the model, providing less resolution for forces and stress to interact in the z direction, and thus less accurate behavior is modelled.

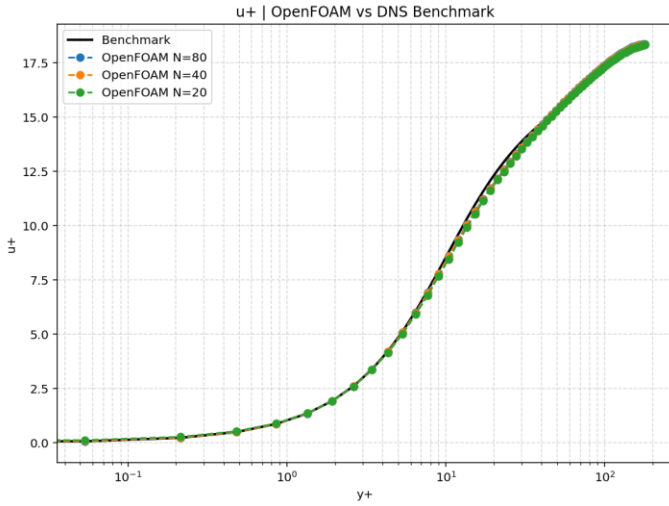


FIGURE 2: U^+ vs y^+ for $N = 20, 40, 80$

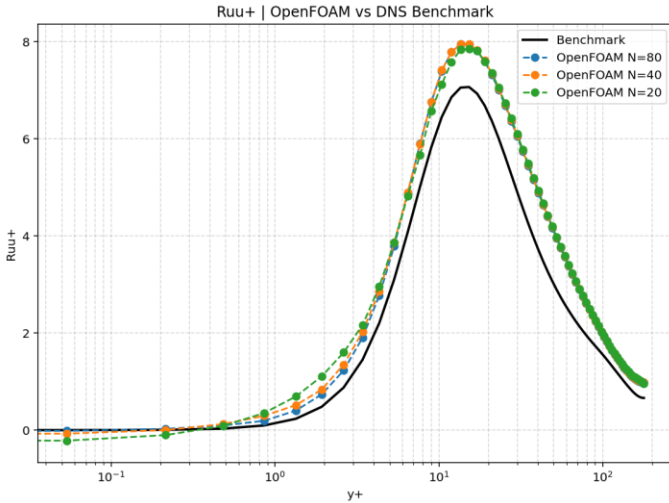


FIGURE 3: R_{uu}^+ vs y^+ for $N = 20, 40, 80$

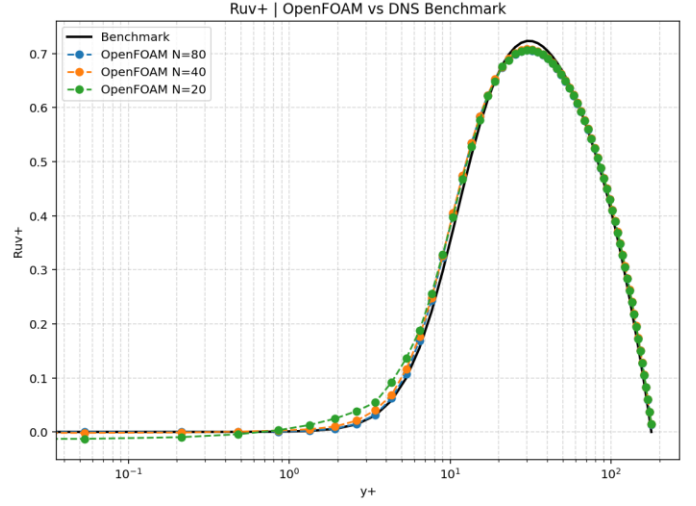


FIGURE 4: R_{uv}^+ vs y^+ for $N = 20, 40, 80$

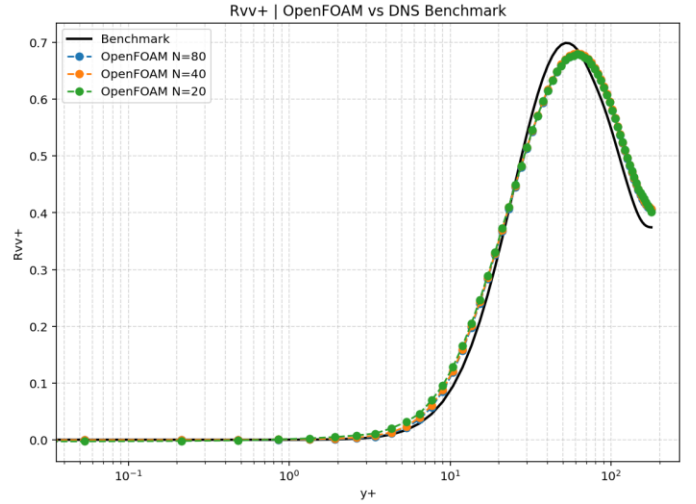


FIGURE 5: R_{vv}^+ vs y^+ for $N = 20, 40, 80$

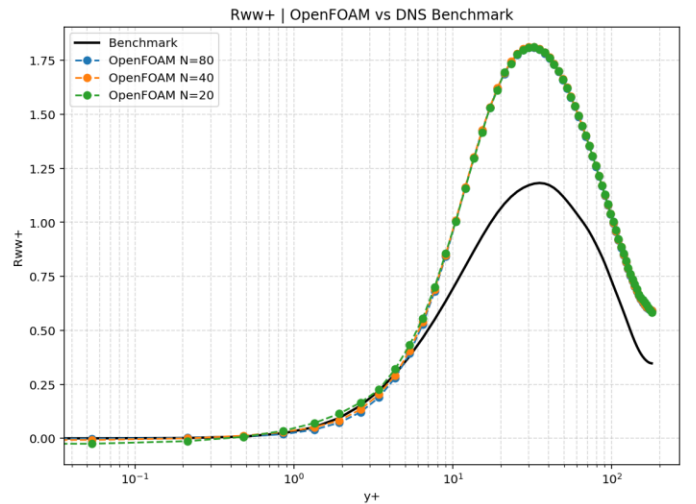


FIGURE 6: R_{ww}^+ vs y^+ for $N = 20, 40, 80$

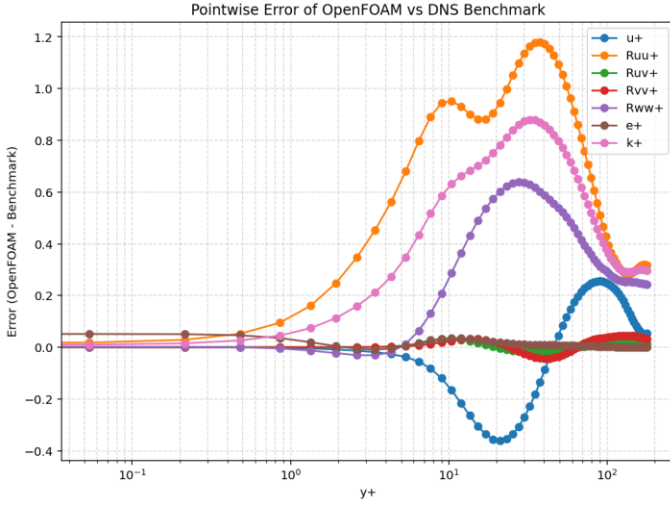


FIGURE 7: Error between DNS and OpenFOAM simulation

TABLE 1: L_2 norms of turbulence quantities for each mesh

Quantity	20 Cells	40 Cells	80 Cells
u^+	1.9165e-01	1.8300e-01	1.8597e-01
R_{uu}^+	7.3814e-01	7.3408e-01	7.1588e-01
R_{uv}^+	1.5234e-02	1.3752e-02	1.2870e-02
R_{vv}^+	3.1454e-02	3.0887e-02	3.0130e-02
R_{ww}^+	3.8111e-01	3.8008e-01	3.7654e-01
ϵ^+	1.3546e-02	1.5249e-02	1.8158e-02
k^+	5.5588e-01	5.5285e-01	5.4177e-01

TABLE 2: L_∞ norms of turbulence quantities for each mesh

Quantity	20 Cells	40 Cells	80 Cells
u^+	4.2971e-01	3.8443e-01	3.6046e-01
R_{uu}^+	1.2131e+00	1.1975e+00	1.1805e+00
R_{uv}^+	3.1793e-02	3.1659e-02	2.8910e-02
R_{vv}^+	4.6695e-02	4.6417e-02	4.4789e-02
R_{ww}^+	6.4555e-01	6.4431e-01	6.3918e-01
ϵ^+	3.6895e-02	3.6671e-02	5.1253e-02
k^+	9.0373e-01	8.9360e-01	8.8026e-01

3.2 Error and Uncertainty

As previously stated, no information regarding experimental uncertainty or input uncertainty was provided by the paper, thus leading to the assumption of the DNS as “truth”. However, a flaw arises in that the only error I have to compare with the comparison error comes from discretization error. This implies that any error between the DNS and the OpenFOAM results (the comparison error) must come from solely discretization error, or model error.

Returning to the tables and Figure 7, the Reynolds normal stress in the u direction has the largest absolute error, though this is only due to the overall magnitude of the turbulence statistic, as both dissipation and R_{ww} have higher relative

error rates. Comparison error was chosen to be reported in terms of absolute as opposed to relative due to the fact that many values of the benchmark data were zero in some places, making normalization impossible for those errors. Keeping comparison error in absolute terms will also make it easier to compare this error with the discretization error.

Discretization error was computed through a GCI method with three mesh resolutions, $N = 20, 40$, and 80 . The results of the GCI analysis are plotted in the form of error bars representing the discretization error in Figures 8 through 12, alongside the benchmark results. These results can also be found in the python file, “proj.py”. It is notable that the OpenFOAM and DNS data for u^+ and R_{uv} are considered within statistical agreement for the vast majority of the points, while R_{uu} , R_{vv} , and R_{ww} only have statistical agreement in the viscous sublayer region. This behavior is somewhat expected, since R_{uv} will benefit the most from a finer discretization in the y direction than the other Reynolds stress term due to having better resolution. Interestingly, each plot shows a maximum discretization error at the boundary between the viscous sublayer and the buffer region. This is most likely due to the fact that the mesh is not uniform, but has a geometric stretch to be finer at the walls, and start to coarsen out as the mesh moves further away. At the buffer region, a y^+ of about 5-10, the mesh starts transitioning to fully uniform in the center. Additionally, since the geometric stretch factor is set and unchanged as N is perturbed, that particular spot will get vastly different mesh size experiences, as it exists between the consistently uniform center and the consistent fine wall mesh.

The observed order of accuracy was calculated for all data sets, but only about 2 terms had enough valid points to make a statement on the observed order of accuracy (POBS). For U^+ , only 6 values near the wall are considered unreliable due to excessively large POBS values (values above 4), out of a total of 65. The consistent values, however, disagree. In the buffer region, the average POBS is 2, but for $y^+ > 30$, the POBS drops to closer to 1. The second data set with a coherent POBS was actually ϵ^+ , which had POBS and discretization error calculated (like all other simulation variables, but just not reported due to time and page length constraints). After the diverging values near the wall fade out, the POBS for dissipation appears to average to 1.

The POBS near the wall for the Reynolds stress terms is estimated to be 1, based on the limited reliable data that was calculated. However, the data quickly transforms into unreliable POBS at about 10 wall units away from the wall. The reasoning for diverging and unreliable POBS data is that there was not monotonic convergence. This is likely due to the fact that finer mesh runs were cut short due to their simulated time reaching 100,000 seconds as opposed to their pressures and velocities reaching a certain tolerance, resulting in inconsistent values. It should be noted that all calculated POBS were positive and above 0.5, suggesting that finer mesh resolutions do in fact grant better convergence, which is expected. All values of POBS for all recorded quantities are available in “proj.py”, located in the “data” folder.

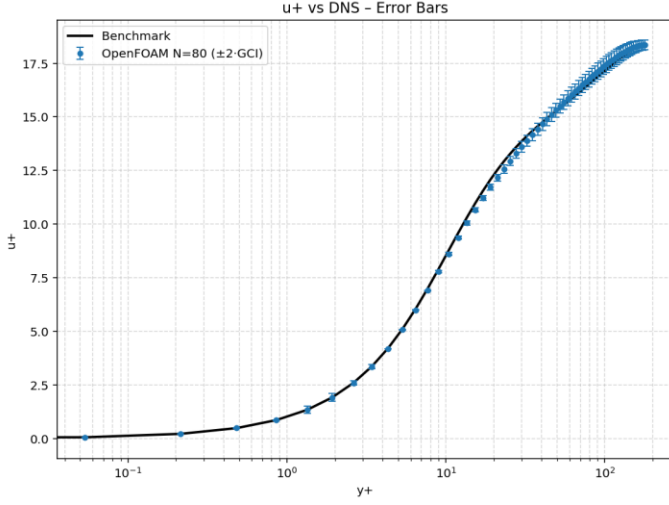


FIGURE 8: U^+ (with discretization error bars) vs y^+ at $N = 80$

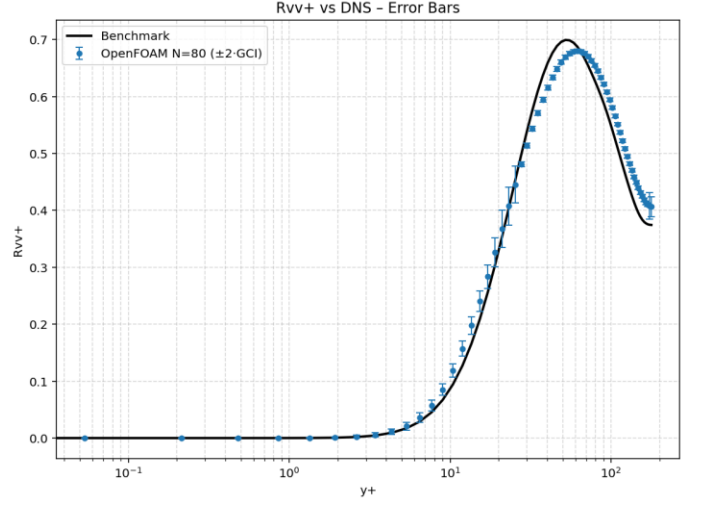


FIGURE 11: R_{vv}^+ (with discretization error bars) vs y^+ at $N = 80$

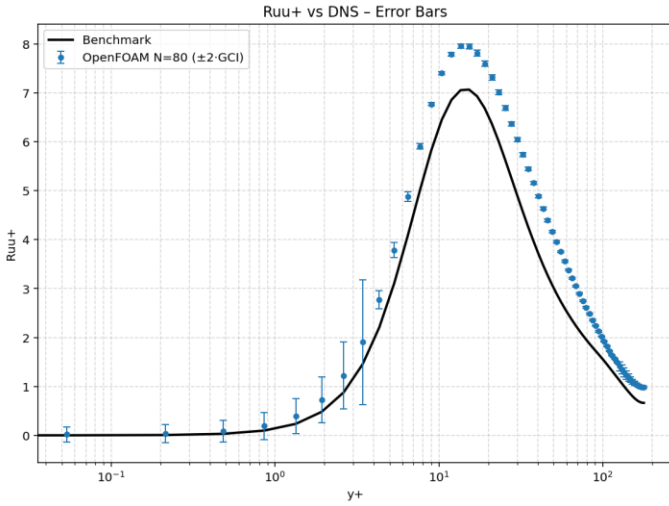


FIGURE 9: R_{uu}^+ (with discretization error bars) vs y^+ at $N = 80$

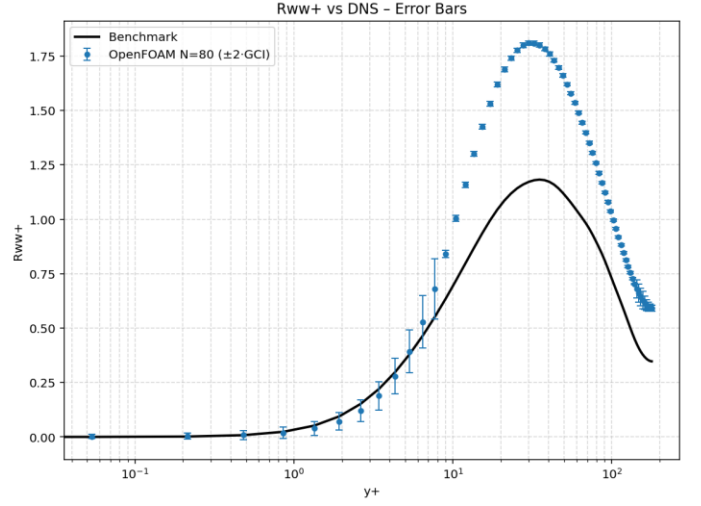


FIGURE 12: R_{ww}^+ (with discretization error bars) vs y^+ at $N = 80$

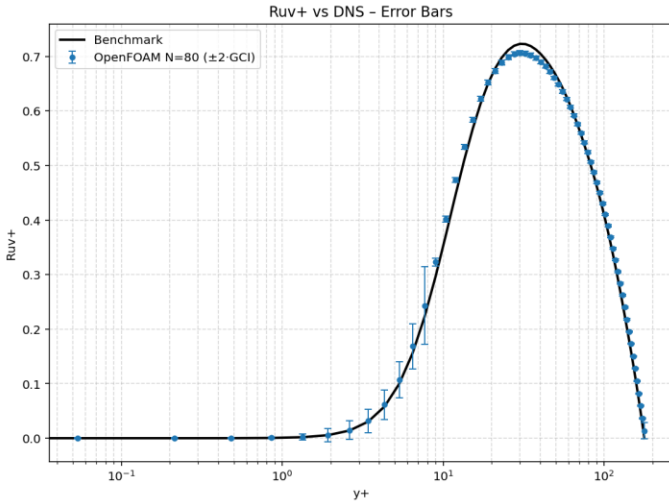


FIGURE 10: R_{uv}^+ (with discretization error bars) vs y^+ at $N = 80$

3.3 Model Error

Finally, the mean values of comparison error and discretization error is shown in Table 3. The only term that has a comparison error larger than discretization error by an order of magnitude is R_{ww}^+ . This was expected, since the modeling choice to simulate 2D flow directly impacts R_{ww}^+ , which depends on spanwise velocity fluctuations which are not being modeled well due to no incoming or outgoing spanwise flow through the side walls. The turbulence terms u^+ and ϵ^+ had discretization errors larger than comparison error, meaning that the chosen model is a decent fit, since its error falls within the statistical noise. Other turbulence terms, such as the rest of the Reynolds stress terms and turbulent kinetic energy had larger comparison errors than discretization errors, though they were of less than an order of magnitude in difference. This suggests that while the model choices for these terms may not be the best, it is not drastically bad enough to say that all the errors come from modelling errors.

TABLE 3: Mean values of comparison error and discretization error for each turbulence variable.

Component	Comparison Error	Discretization Error
k^+	4.7522e-01	6.5514e-02
ϵ^+	1.1608e-02	2.0030e-02
R_{ww}^+	3.2089e-01	2.0410e-02
R_{vv}^+	2.5500e-02	6.6170e-03
R_{uv}^+	1.0700e-02	6.0529e-03
R_{uu}^+	6.2011e-01	9.3168e-02
u^+	1.5190e-01	2.3386e-01

4. CONCLUSION

In summary, a validation study of fully developed plane channel flow at $Re \approx 180$ was performed using an OpenFOAM EBRSM and compared against DNS data. The GCI analysis showed that discretization errors were generally small with most discrepancies instead arising from modeling limitations, though some quantities exhibited unstable observed orders of accuracy. Variables such as u^+ and R_{uv}^+ showed good agreement with DNS, while the Reynolds normal stresses deviated more significantly, especially R_{ww}^+ due to the lack of spanwise fluctuations in the 2D setup. Overall, the results indicate that the chosen RANS approach captures mean flow behavior reasonably well but is limited in predicting anisotropic turbulence quantities.

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Thank you to the TAMU HPC for providing cpu cores to run more cases. The writer reports the use of AI purely for the purposes of grammar and spell checking – all original thoughts are their own (Prompt: “Please check this paper for proper grammar and spell checking. Do not rewrite anything”).

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